AJ-ORBIT

BLUETOOTH RC CAR · COMPONENTS CHECKLIST

Build Beyond Limits

Project overview

A four-wheel Bluetooth-controlled robot car built around an Arduino Uno and an L293D four-motor shield. This list assumes a shield with M1, M2, M3 and M4 motor ports and a compatible motor-control library.

1. Main components

NO.	COMPONENT	QTY.	PURPOSE / NOTES
1	**Arduino Uno**	1	Main controller board
2	**L293D four-motor driver shield**	1	Must be compatible with your shield design; confirm its library before uploading
3	**HC-05 or HC-06 Bluetooth module**	1	Receives driving commands from a phone
4	**DC geared motors**	4	Check motor voltage and stall current against the driver's limits
5	**Robot car chassis**	1	Frame to mount the board, motors and battery
6	**Wheels**	4	Match the motor shafts and chassis
7	**Battery pack + holder**	1	Voltage must suit motors and shield; do not power motors from the Uno 5 V pin
8	**Power switch**	1	Rated for the battery and motor current
9	**Jumper wires**	As needed	For Bluetooth and any required signal/power connections
10	**USB cable for Arduino Uno**	1	Uploading code and serial debugging

2. Small items and optional parts

ITEM	QTY.	NOTES
HC-05 RX voltage divider (e.g., two resistors)	1 set	Use if required by your Bluetooth breakout; protect the module's 3.3 V RX logic input
Screws, nuts, spacers and cable ties	As needed	Secure the boards and prevent loose wires
Caster wheel / support wheel	Optional	Usually unnecessary for a four-wheel chassis
Multimeter	Optional	Helpful for checking polarity and supply voltage

3. Before powering on

- ■ Confirm your exact L293D shield model and its library requirements.
- ■ Check the battery voltage and the motors' stall current against the shield ratings.
- ■ Connect all grounds as required by the shield design; follow its printed power labels.
- ■ Test the car with its wheels lifted and keep fingers clear of moving parts.

■ Disconnect motor power before wiring or plugging/unplugging the shield.

> **Library note:** The AFMotor library is needed only if your shield is a compatible V1-style design. It is not embedded inside the .ino file; include the official library link and installation steps in your GitHub README.